Multiphysics Benchmark Results from Moltres

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INTRODUCTION

Molten Salt Reactors (MSRs) have attracted increasing research interest over the past two decades. Reactor analysis software are essential tools in reactor development because they help inform design choices in line with overarching goals such as maximizing safety, reducing proliferation risks, and improving fuel efficiency.

Circulating fuel MSRs present new challenges in computational reactor safety analysis arising from the liquid fuel form. In the primary coolant holding the dissolved fissile material, the strong negative reactivity feedback causes strong and near-instantaneous interactions between reactor power and thermal-hydraulics. Thus, unexpected changes in coolant flow and temperature have a strong impact on reactor power and vice versa.

This paper presents benchmarking results from Moltres, a coupled neutronics/thermal-hydraulics simulation software for MSRs. Moltres is built on the Multiphysics Object-Oriented Simulation Environment (MOOSE) [1] finite element framework, which provides various tools for performing fully-coupled, NEWTON-based solves. The benchmark, referred to as the *CNRS benchmark* [2], is a numerical benchmark for software dedicated to modeling MSRs.

BACKGROUND & METHODOLOGY

CNRS Benchmark

The CNRS benchmark [2] consists of three *Phases* and eight *Steps* in total. Each step is a well-defined subproblem for systematically assessing MSR software capabilities and pinpointing the source of discrepancies between software. Phase 0 consists of single-physics problems in fluid dynamics, neutronics, and temperature, respectively. Phase 1 consists of coupled, steady-state eigenvalue problems. Lastly, Phase 2 consists of a coupled, transient scenario. Table I provides brief descriptions of each step, except for the last transient case as it is still a work in progress at the time of writing.

The domain geometry is a 2m by 2m square cavity filled with LiF-BeF_2-UF_4 molten salt at an initial temperature of 900K [2]. Standard vacuum boundary conditions apply for neutron flux along the boundaries, with homogeneous boundary conditions for delayed neutron precursors. No-slip boundary conditions apply for flow in the cavity, except the top boundary when the subproblem imposes forced convection via lid-driven cavity flow. For temperature, all boundaries are insulated and we simulate salt cooling with the following volumetric heat sink equation:

$$q'''(\vec{r}) = \gamma\,(900 - T(\vec{r}))\,, \tag{1}$$

where

q''' = volumetric heat sink [W·m^{-3}],
γ = heat transfer coefficient [W·m^{-3}·K^{-1}],
$T(\vec{r})$ = temperature at point ($\vec{r}$) [K].

Tiberga et al. [2] used Serpent 2 [3] with the JEFF-3.1 library [4] to generate multigroup neutronics data for the LiF-BeF_2-UF_4 salt in the domain at 900K. They condensed the neutronics data into six energy groups and eight precursor groups. Their paper contains all relevant data [2]. The benchmark prescribes the following equations to govern the temperature dependence in the cross sections and the neutron diffusion coefficients:

$$\Sigma_i(T) = \Sigma_i(T_{ref})\frac{\rho_{fuel}(T)}{\rho_{fuel}(T_{ref})}, \tag{2}$$

$$D(T) = D(T_{ref})\frac{\rho_{fuel}(T_{ref})}{\rho_{fuel}(T)}, \tag{3}$$

where

Σ_i = relevant macroscopic cross section [cm−1],
D = neutron diffusion coefficient [cm^2·s^{-1}],
ρ_{fuel} = density of the fuel salt [kg·m^{-3}],
T_{ref} = reference temperature = 900 K.

The benchmark also prescribes for incompressible Navier-Stokes and the Boussinesq approximation for buoyancy when evaluating the salt flow in the domain but places no restrictions on the type of neutronics model.

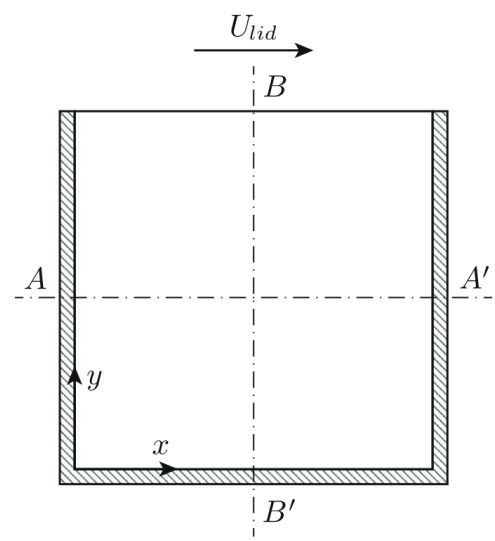

Fig. 1: The CNRS benchmark problem domain. The benchmark compares pointwise observable data along the centerlines AA' and BB'. Retrieved from [2].

TABLE I: Description of each benchmark subproblem.

Step	Description	Input parameters	Observables
Step 0.1: Velocity field	Solve for the steady-state incompressible flow solution from lid-driven cavity flow.	$U_{lid} = 0.5$ m·s^{-1}	(u_x, u_y) along AA' and BB';
Step 0.2: Neutronics	Solve for the fission rate density $\sum_i^G \Sigma_{f,i}\phi_i(\vec{r})$ and reactivity ρ in a static, isothermal fuel configuration.	$U_{lid} = 0$ m·s^{-1} $T = 900$ K $P = 1$ GW	$\sum_i^G \Sigma_{f,i}\phi_i(\vec{r})$ along AA'; ρ;
Step 0.3: Temperature	Solve for the temperature distribution T arising from the velocity field $(u_x, u_y)_{s_{0.1}}$ and fission heat generation $Q_{s_{0.2}}$ from Steps 0.1 and 0.2, respectively, and a volumetric heat sink (Equation 1).	$(u_x, u_y) = (u_x, u_y)_{s_{0.1}}$ $Q = Q_{s_{0.2}}$ $\gamma = 10^6$ W·m^{-3}·K^{-1}	T along AA' and BB';
Step 1.1: Circulating fuel	Solve for the delayed neutron source distribution $\sum_j \lambda_j C_j$ and reactivity ρ in a non-static, isothermal fuel configuration.	$(u_x, u_y) = (u_x, u_y)_{s_{0.1}}$ $T = 900$ K $P = 1$ GW	$\sum_j \lambda_j C_j$ along AA' and BB'; $\rho - \rho_{s_{0.2}}$;
Step 1.2: Power coupling	Solve for the temperature distribution T, reactivity ρ, and fission rate density $\sum_i^G \Sigma_{f,i}\phi_i(\vec{r})$ under the fixed velocity field from Step 0.1.	$(u_x, u_y) = (u_x, u_y)_{s_{0.1}}$ $P = 1$ GW $\gamma = 10^6$ W·m^{-3}·K^{-1}	T along AA' and BB'; $\rho - \rho_{s_{1.1}}$; $\sum_i^G \Sigma_{f,i}\phi_i(\vec{r}) - \left[\sum_i^G \Sigma_{f,i}\phi_i(\vec{r})\right]_{s_{0.2}}$;
Step 1.3: Buoyancy	Solve for the velocity components (u_x, u_y), temperature distribution T, delayed neutron source distribution $\sum_j \lambda_j C_j$, and reactivity ρ in a fully coupled system.	$P = 1$ GW $U_{lid} = 0$ m·s^{-1} $\gamma = 10^6$ W·m^{-3}·K^{-1}	(u_x, u_y) along AA' and BB'; T along AA' and BB'; $\sum_j \lambda_j C_j$ along AA' and BB'; $\rho - \rho_{s_{0.2}}$;
Step 1.4: Full coupling	Solve for the reactivity ρ in a fully coupled system with external momentum-driven flow and buoyancy effects and various permutations of P and U_{lid} values.	$\gamma = 10^6$ W·m^{-3}·K^{-1} $P = [0, 1]$ GW $U_{lid} = [0, 0.5]$ m·s^{-1}	$\rho - \rho_{s_{0.2}}$;

Moltres

Moltres [5] is an open-source, MOOSE-based application designed for multi-physics simulations of MSRs. Moltres depends on the MOOSE finite element framework for its meshing and parallel, nonlinear NEWTON-based solver capabilities, and standard physics modules such as the Navier-Stokes module [6] in MOOSE.

Moltres solves the multigroup neutron diffusion equation for an arbitrary number of energy and precursor groups. For this study, we coupled Moltres' neutronics capabilities with MOOSE's incompressible Navier-Stokes capabilities [6] to model precursor drift and thermal-hydraulics.

For this study, we discretized the problem domain into a 200 by 200 structured mesh, resulting in uniform mesh elements of dimensions 1cm by 1cm. We approximated most of the relevant variables, i.e., neutron fluxes, velocity components, pressure, and temperature, using first-order Lagrange shape functions. The only exception is the precursor concentration variables, which we approximated using zeroth-order monomial shape functions and solved using the Discontinuous Galerkin finite element method to eliminate spurious numerical oscillations from the high Schmidt number flow. As for the high Reynolds number flow, we stabilized the incompressible Navier-Stokes equations using the streamline upwind Petrov-Galerkin and pressure-stabilizing Petrov-Galerkin stabilization methods [6] provided in MOOSE.

RESULTS

In this section, we compare the results from Moltres for each benchmark step to the results reported in the the benchmark paper [2]. The benchmark paper contains results from four different MSR simulation software packages. The authors of the benchmark performed code-to-code verification by reporting the average discrepancy ϵ from all software packages for each benchmark step. The discrepancy ϵ_c from each software package for each measured quantity Q_c calculated using the following equation:

$$\epsilon_c = \sqrt{\frac{\sum_{i=1}^{N_p}\left(Q_c(\vec{r_i}) - Q_{avg}(\vec{r_i})\right)^2}{\sum_{i=1}^{N_p} Q_{avg}^2(\vec{r_i})}} \tag{4}$$

where

$$\begin{aligned} N_p &= 201 \\ &= \text{number of sampling points of quantity } Q, \\ N_c &= \text{number of MSR software packages,} \\ Q_{avg}(\vec{r_i}) &= \frac{1}{N_c}\sum_{c=1}^{N_c} Q_c(\vec{r_i}) \\ &= \text{average value of } Q \text{ at } \vec{r_i} \text{ from all packages.} \end{aligned}$$

TABLE II: Discrepancy values for the results from Phase 0.

Step	Observable	Source	Discrepancies [%] Along AA'	Discrepancies [%] Along BB'
0.1	u_x [m·s^{-1}]	Moltres	0.247	0.266
		Benchmark	0.253	0.318
	u_y [m·s^{-1}]	Moltres	0.540	0.467
		Benchmark	0.598	0.795
0.2	$\sum_i^G \Sigma_{f,i}\phi_i(\vec{r})$ [m^{-3}·s^{-1}]	Moltres	0.313	N.A.
		Benchmark	0.285	N.A.
0.3	T [K]	Moltres	0.090	0.164
		Benchmark	0.085	0.083

TABLE III: Reactivity ρ and change in reactivity $(\rho_a - \rho_b)$ values from Steps 0.2, 1.1, 1.2, and 1.3. All units are in pcm.

Source	Step 0.2 $\rho_{s_{0.2}}$	Step 1.1 $\rho_{s_{1.1}} - \rho_{s_{0.2}}$	Step 1.2 $\rho_{s_{1.2}} - \rho_{s_{1.1}}$	Step 1.3 $\rho_{s_{1.2}} - \rho_{s_{0.2}}$
Moltres	465.6	−62.9	−1142.2	−1207.7
CNRS-SP_1	411.3	−62.5	−1152.0	−1220.5
CNRS-SP_3	353.7	−62.6	−1152.7	−1220.7
Polimi	421.2	−62.0	−1161.0	−1227.0
PSI	411.7	−63.0	−1154.8	−1219.6
TUD-S_2	482.6	−62.0	−1145.2	−1208.5
TUD-S_6	578.1	−60.7	−1122.0	−1184.4

Phase 0 results & discussion

Table II shows the discrepancy values from Moltres for Phase 0 and the corresponding average discrepancy values from the benchmark [2]. Moltres performs excellently as all discrepancy values are within the same order of magnitude as those from the benchmark, except the discrepancy value for T along centerline BB'. We also observe in table III that the ρ value from Moltres falls within the range of ρ values from the benchmark.

Moltres reports a slightly larger discrepancy in the fission rate density as we used zeroth-order shape functions to approximate the precursor concentrations C_i. These shape functions perform worse in approximating C_i near the boundaries where the gradient is steeper in the homogeneous domain with no reflectors.

The temperature values at the end of centerline BB', near the top boundary, deviated significantly (912 K from Moltres vs. 935 K from the benchmark). The finite element method struggles with the velocity boundary condition discontinuity on the top left corner of the domain. As such, we observe the temperature deviation directly downstream of the discontinuity.

Phase 1 results & discussion

Figure 2 shows the typical temperature distribution and velocity streamlines from running a fully coupled problem in Step 1.4. Table IV shows the discrepancy values from Moltres for Steps 1.1, 1.2, and 1.3, and the corresponding average discrepancy values from the benchmark [2]. The subsequent subsections discuss the results for each benchmark step.

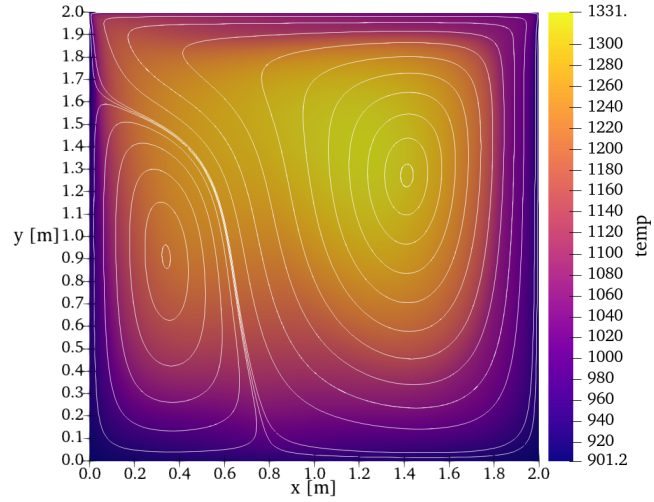

Fig. 2: The temperature distribution in the domain for the fully coupled system (Step 1.4) with buoyancy effects, P = 1 GW, and U_{lid} = 0.5 m·s^{-1}. The lines correspond to the streamlines of the velocity field.

Step 1.1: Circulating fuel

Moltres performs relatively worse than the benchmark average, but the discrepancies are still on the same order of magnitude. We attribute this to the same zeroth-order shape function approximation for the precursor concentration mentioned in Step 0.2.

On the other hand, we observe in Table III that the change in ρ relative to Step 0.2 falls within the reported range of ρ values from the benchmark.

TABLE IV: Discrepancy values for the results from Phase 1.

Step	Observable	Source	Discrepancies [%] Along AA'	Discrepancies [%] Along BB'
1.1	$\sum_j \lambda_j C_j$ [m^{-3}·s^{-1}]	Moltres	0.606	0.378
		Benchmark	0.346	0.294
1.2	T [K]	Moltres	0.076	0.179
		Benchmark	0.095	0.089
	$\Delta\left[\sum_i^G \Sigma_{f,i}\phi_i(\vec{r})\right]_{s_{1.2}-s_{0.2}}$ [m^{-3}·s^{-1}]	Moltres	1.110	1.082
		Benchmark	1.576	1.133
1.3	u_x [m·s^{-1}]	Moltres	0.123	N.A.
		Benchmark	0.691	N.A.
	u_y [m·s^{-1}]	Moltres	0.236	0.238
		Benchmark	0.329	0.356
	T [K]	Moltres	0.064	0.070
		Benchmark	0.057	0.080
	$\sum_j \lambda_j C_j$ [m^{-3}·s^{-1}]	Moltres	1.079	0.5780
		Benchmark	0.460	1.194

Step 1.2: Power coupling

For this step, Moltres performed better than the benchmark average except the temperature distribution along BB',

the cause of which we addressed in Step 0.3. The smaller discrepancy in the change in fission rate density indicates that Moltres is on average more self-consistent since this quantity measures the difference between two sets of data from the same software. From Table III, Moltres also reports a change in ρ, relative to Step 1.1, that falls within the range of benchmark values.

Step 1.3: Buoyancy

Moltres performs reasonably well in Step 1.3, with smaller discrepancies in most of the velocity components and temperature distribution than the benchmark average. Moltres bucks the trend for discrepancies in the delayed neutron source as the discrepancy along AA' is noticeably worse than the benchmark average, but Moltres performs relatively better along BB'. In Table III, we observe once again that the change in ρ in Moltres falls within the range of benchmark values.

Step 1.4: Full coupling

Table V shows the change in ρ under the various U_{lid} and P values. We refer readers to Tiberga et al.'s paper [2] for the benchmark's corresponding values. The results from Moltres all fall within the range of benchmark values for all cases. Furthermore, the $\Delta\rho$ values are all within 1.1 pcm of the corresponding values from the TUD-S_2 model in the benchmark paper. Given that the S_2 discrete ordinates method is qualitatively similar to the multigroup neutron diffusion method, this shows that Moltres is consistent to the benchmark within the limitations brought about by the neutron diffusion method.

TABLE V: Reactivity change in Step 1.4, relative to Step 0.2 under various U_{lid} and P values.

	$\rho - \rho_{s0.2}$ [pcm]				
U_{lid} [m·s^{-1}] \ P [GW]	0.2	0.4	0.6	0.8	1.0
0.0	-263.7	-498.3	-730.9	-966.7	-1207.7
0.1	-265.9	-498.7	-730.6	-966.0	-1206.7
0.2	-268.1	-498.8	-729.4	-963.7	-1203.6
0.3	-269.9	-498.5	-727.8	-960.8	-1199.5
0.4	-271.9	-498.5	-726.5	-958.3	-1195.7
0.5	-274.2	-498.7	-725.6	-956.4	-1192.7

CONCLUSIONS

In this study, we performed code-to-code verification on Moltres based on the CNRS numerical benchmark [2] for multi-physics, MSR simulation software. The benchmark assesses simulation software through subproblems involving single-physics and coupled neutronics/thermal-hydraulics.

The results showed that Moltres is mostly consistent with the other software presented in the benchmark paper. The percentage discrepancies in the various neutronics, velocity, and temperature quantities mostly fall within the same order of magnitude. Minor deviations in the temperature in Steps 0.3 and 1.2 stem from the discontinuous velocity boundaries on the top corners in the lid-driven cavity flow, while minor deviations in the flux and delayed neutron source in Steps 0.2, 0.3, and 1.1 stem from the zeroth-order shape function approximation on the precursor concentration. The results from Moltres agree closest to the TUD-S_2 model from the benchmark, which implements the S_2 discrete ordinates method for neutron transport.

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REFERENCES

1. D. R. GASTON, C. J. PERMANN, J. W. PETERSON, A. E. SLAUGHTER, D. ANDRŠ, Y. WANG, M. P. SHORT, D. M. PEREZ, M. R. TONKS, J. ORTENSI, L. ZOU, and R. C. MARTINEAU, "Physics-based multiscale coupling for full core nuclear reactor simulation," *Annals of Nuclear Energy*, **84**, 45–54 (Oct. 2015).
2. M. TIBERGA, R. G. G. DE OLIVEIRA, E. CERVI, J. A. BLANCO, S. LORENZI, M. AUFIERO, D. LATHOUWERS, and P. RUBIOLO, "Results from a multi-physics numerical benchmark for codes dedicated to molten salt fast reactors," *Annals of Nuclear Energy*, **142**, 107428 (Jul. 2020).
3. J. LEPPANEN, M. PUSA, T. VIITANEN, V. VALTAVIRTA, and T. KALTIAISENAHO, "The Serpent Monte Carlo code: Status, development and applications in 2013," *Annals of Nuclear Energy*, **82**, 142–150 (Aug. 2014).
4. A. KONING, R. FORREST, M. KELLETT, R. MILLS, H. HENRIKSSON, and Y. RUGAMA, "The JEFF-3.1 nuclear data library: JEFF report 21," Tech. Rep. 21, OECD Nuclear Energy Agency (2006), oCLC: 1039431841.
5. A. LINDSAY, G. RIDLEY, A. RYKHLEVSKII, and K. HUFF, "Introduction to Moltres: An application for simulation of Molten Salt Reactors," *Annals of Nuclear Energy*, **114**, 530–540 (Apr. 2018).
6. J. W. PETERSON, A. D. LINDSAY, and F. KONG, "Overview of the Incompressible Navier-Stokes simulation capabilities in the MOOSE Framework," *arXiv:1710.08898 [math]* (Oct. 2017), arXiv: 1710.08898.